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Guidelines for Project Proposals

Mechanical Project 478
Mechatronic Project 478
Mechatronic Project 488

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2019

saam vorentoe · masiye phambili · forward together

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1. Introduction

This document, which includes an example of a project proposal, serves as a guideline for the preparation of project proposals in Mechanical Project 478, Mechatronic Project 478 and Mechatronic Project 488. Please note that the content of a specific project proposal will depend on the project's topic and, therefore, the example serves as a guideline only.

The point of departure for the project proposal is that the proposal should "sell" the project to a potential client. Students should consider it to be a situation where they have to compete with their classmates to be awarded a part of a limited source of funds to realise the project. The project proposal has to, in other words, explain the goals of the project, that the project is significant and that the proposal presents a workable plan to realise the project in a meaningful way. Ertas & Jones [1996:426] describes the purpose of a project proposal as follows:

> *The purpose of submitting a proposal is to put forth an idea or solution to a need for which the potential customer is known or thought to be interested. Since the purpose is to sell an idea or solution, the proposal must first* ***establish that the need exists*** *and then provide information adequate to convince the potential customer that* ***the solution proposed is the right one*** *and that the* ***proposer is capable*** *of accomplishing the effort described within the planned schedule and budget.*

2. Instructions

An example, in the recommended format for project proposals, is attached. The usual report format, as described in the "Guide for writing Technical Reports" by AH Basson and TW von Backström, must be followed for the project proposal. Some particular aspects are explained in the remainder of this section.

The format of the executive summary, which is based on Heilmeier's Catechism, has to be strictly adhered to. The executive summary may only comprise one page.

The Introduction section must give the background of the project. As is customary, it should start with the "big picture" and gradually narrow the perspective to the current project, to show how the proposed project relates to the "big picture". Preceding and concurrent related projects should also be mentioned.

The Objectives section should concisely state the project's objectives, taking care not to include the steps required to achieve the objectives as objectives themselves. The objectives would typically be formulated as results that would have direct value to the project's client (i.e. the person or company paying for the project). The formulation of the objectives should also take into account that, at the end of the project, the examiners will assess to what extent the student has achieved the objectives.

The Motivation section should focus on why it is worth pursuing the specific objectives of the project, or what value would a client get from investing in the project. The Motivation section should avoid attempting to justify the background to the project.

The Planned Activities section should reflect the objectives and scope of the specific project. It should be clear from the description of each activity what value the activity would have or, in other words, how that activity contributes towards achieving the project's objectives. The activities should be refined sufficiently to assist in compiling a time schedule (in the Gantt chart), which should show which activities can be conducted in parallel and which activities have to be done in a sequence.

A budget must also be compiled for the project as part of the proposal. Record must be kept of the actual costs incurred during the execution of the project, since the actual costs must be given in the final report, as well as a techno-economic evaluation. Guidelines for compiling the budget are:

- The budget can be based on a nominal tariff of R450 per hour for junior engineers (for the student's own work). The number of engineering hours in the budget should correspond to the number of credits, with 10 hours of work per credit.
- "Running Costs" are costs of services or goods that will be “consumed” and not included elsewhere in the budget, e.g. travel expenses, copying, printing, lubricants, bought-out items, etc.
- "Facility Use" makes provision for the depreciation and maintenance of equipment that was purchased previously or the infrastructure used, e.g. laboratory instrumentation, computers and wind tunnels. Depreciation is usually calculated on the basis of the replacement value of the equipment, the frequency of use, and the expected lifetime. In the project proposal an amount of R50 per hour per R100 000 replacement value can be used.
- "Capital Costs" are expenses undertaken for purchasing equipment specifically for the completion of the project, but only where the equipment will still be available for other uses after the project has been completed, e.g. computers and instrumentation. At Stellenbosch University only items with purchase costs exceeding R2000 are considered to be "assets" and would be included in the capital cost budget.
- "MMW Labour" is the time that the technicians from the Mechanical and Mechatronic Workshop spend on the project and can be budgeted for at R300 per hour.

- "MMW Material" is for the direct running expenses (typically material costs) that MMW will incur for the work on the project.

All expenses incurred outside the university (typically running, capital and material) should include VAT.

The section on the Project Risk Assessment should consider the risks that could prevent the project from being successfully completed. Routine health and safety risks during the execution of the project need not be listed. Measures that will be used to mitigate the project risks must be described in this section.

3. Conclusion

These guidelines and the example will help students doing final year projects in the Mechanical Engineering or Mechatronic Engineering programmes to plan their projects. Planning is an important first step towards successfully executing a project.

4. References

Ertas, A, Jones, JC "The Engineering Design Process", 2nd Ed, John Wiley & Sons, New York, 1996.

Appendix A: Project Proposal Example

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THE TECHNICAL FEASIBILITY OF A MULTIPURPOSE ELECTRICAL VEHICLE PROPULSION SYSTEM

Mechatronic Project 478
Project Proposal

Mr HAR de Werker
Student no 12345678

Supervisor: Dr I Knowitall

24 March 2019

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Declaration

I have read and understand the Stellenbosch University Policy on Plagiarism and the definitions of plagiarism and self-plagiarism contained in the Policy [Plagiarism: The use of the ideas or material of others without acknowledgement, or the re-use of one's own previously evaluated or published material without acknowledgement or indication thereof (self-plagiarism or text-recycling)].

I also understand that direct translations are plagiarism, unless accompanied by an appropriate acknowledgement of the source. I also know that verbatim copy that has not been explicitly indicated as such, is plagiarism.

I know that plagiarism is a punishable offence and may be referred to the University's Central Disciplinary Committee (CDC) who has the authority to expel me for such an offence.

I know that plagiarism is harmful for the academic environment and that it has a negative impact on any profession.

Accordingly all quotations and contributions from any source whatsoever (including the internet) have been cited fully (acknowledged); further, all verbatim copies have been expressly indicated as such (e.g. through quotation marks) and the sources are cited fully.

I declare that, except where a source has been cited, the work contained in this assignment is my own work and that I have not previously (in its entirety or in part) submitted it for grading in this module/assignment or another module/assignment.

I declare that have not allowed, and will not allow, anyone to use my work (in paper, graphics, electronic, verbal or any other format) with the intention of passing it off as his/her own work.

I know that a mark of zero may be awarded to assignments with plagiarism and also that no opportunity be given to submit an improved assignment.

Name: HAR de Werker

Student no: 12345678

Signature: ....

Date: 24 March 2019

Executive Summary

Title of Project
The technical feasibility of a multipurpose electrical vehicle propulsion system
Objectives
Identification of the propulsion system configuration that will have the best technical feasibility and determining the technical feasibility of producing the selected propulsion system.
What is current practice and what are its limitations?
Current electrical vehicles are single-purpose, typically public transport, goods transport or private road transport.
What is new in this project?
The project will consider multiple applications and evaluate the use of advanced control systems to adapt the propulsion system according to the current use.
If the project is successful, how will it make a difference?
The project will determine whether the expenses involved in a study of the economic viability should be incurred.
What are the risks to the project being a success? Why is it expected to be successful?
Control systems may not be able to deliver an acceptable range of application for the propulsion system. Further, the project may exceed the allowed time and cost. The risks are mitigated by the planning and the resources available.
What contributions have/will other students made/make?
A previous student investigated the state of currently available battery technology.
Which aspects of the project will carry on after completion and why?
An economic feasibility study will be conducted.
What arrangements have been/will be made to expedite continuation?
The chosen configuration will be defined and documented in sufficient detail that the production demands and marketing potential can be determined.

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1. Introduction

There is growing interest in electrical vehicles due to increased concern about climate change. Many companies already market electrical vehicles. Future increases in the price of crude oil will lead to marked increases in the price of petrol and diesel. Although electricity cost has increased significantly in recent years, electricity still remains an economically attractive energy source for vehicles. Further, there have been significant advances in battery technology over the last decade, while internal combustion engines still have shorter life spans and higher maintenance costs than electrical motors. These factors can result in electrical vehicles being able to compete on a lifetime cost basis with petrol and diesel vehicles.

The project proposed here is a technical feasibility study of a propulsion system that will be suitable for a range of electrical motor vehicle applications. Economic considerations will play an obvious part in the study, but the primary focus of this study is the technical aspects.

This project, which Mr HAR de Worker is doing as part of Mechanical Project 448, stems from a proposal by Dr MY Lecturer. The project builds upon a project by Formerstudent [1993] who investigated battery technologies suitable for electrical vehicles. The work forms part of a bigger project aimed at the application of renewable energy and is funded by GreenEnergy Ltd.

This document explains the project's objectives, motivation and planning. The steps that are planned in this study, as well as the expected costs and time scales for the study, are also outlined.

2. Objectives

As mentioned above, this project is aimed at determining the technical feasibility of a multi-purpose electrical vehicle propulsion system. The objectives of the project are therefore:

2.1. Identify the propulsion system configuration that will have the best technical feasibility.

2.2. Prove the technical feasibility of producing the chosen propulsion system.

3. Motivation

Since production volumes in South Africa's vehicle market are fairly small, the economic viability of an electrical vehicle will be increased significantly by using the same electrical propulsion system (battery, electrical distribution, propulsion and control) for a number of different vehicles. One would expect that the body and interior of an electrical vehicle would not differ much from that of a conventional vehicle. It is therefore not deemed necessary to include that in this study.

The Department of Mechanical and Mechatronic Engineering and the Department of Electrical and Electronic Engineering have wide ranging expertise that will be used in this study. These fields of expertise include: the mechanical requirements that a propulsion system should comply with, the development and application of the electrical subsystems, and the development and utilisation of sophisticated electronic control systems.

4. Planned Activities

The following activities are anticipated for the execution of the project. The costs and time scales associated with each activity are given in Appendix A.

4.1. Review Literature

Study the literature on electrical vehicle propulsion to determine the current level of technology and to identify existing propulsion systems.

4.2. Identify Compatible Applications

Investigate possible/probable applications of electrical propulsion systems. Determine the categories of applications (from the perspective of the demands placed on the propulsion system). Determine which applications will be suitable for a multi-purpose propulsion system.

4.3. Compile Design Requirements

Determine the requirements for the propulsion system to be suitable for the chosen combination of applications. Compile a list of the evaluation criteria to be used to compare the different concepts. The whole life cycle of the propulsion system will be considered, including development and testing, manufacturing, operation, support and disposal.

4.4. Investigate Concepts for the Propulsion System

Formulate a set of concepts for the propulsion system based on these ideas, as well as the information gained from the literature study. Investigate each concept using simple analyses to determine the extent to which it meets the criteria set in Activity 4.3. Compile a shortlist of the most promising concepts on these grounds.

4.5. Simulate the Most Promising Concepts

Investigate the shortlisted concepts in detail. Develop a vehicle dynamic simulation of each concept to determine quantitatively the extent to which it meets the design requirements.

4.6. Select Preferred Concept

Use the information gained from the previous activity to make a fully motivated choice of the most suitable concept(s). Throughout, the whole life cycle of the propulsion system will be considered, including development and testing, manufacturing, operation, support and disposal.

4.7. Design Review

Present the findings of the preceding activities to the client and technical advisors to obtain their approval for the selected concept.

4.8. Design Concept Demonstrator

Design a model propulsion system where the core elements of the simulation are presented. The model will be suitable to verify the main results of the simulation. Plan the test procedure that will be used.

4.9. Manufacture Concept Demonstrator

Have the concept demonstrator manufactured by MMW, and supervise the manufacturing.

4.10. Test Concept Demonstrator

Instrument and test the concept demonstrator. Process the test results.

4.11. Validate Simulation

Compare the test results with the results from the simulation.

4.12. Finalise Report

Document the whole investigation, including the preparatory study, the investigative procedures and the results obtained in each activity. Make a recommendation regarding the technical feasibility of a multi-purpose electrical vehicle propulsion system.

5. Project Risk Assessment

This section considers the significant risks that could prevent the project from being completed.

The main risk to the project is that it may not be technically feasible to devise a propulsion system that meets the design requirements and, in particular, that no concept suitable for a sufficient number of different roles is found. In that case, the resulting propulsion system will not be "multipurpose". It is expected that the control system will be the key to making multipurpose application possible. This project risk is therefore primarily related to the control system not being able to deliver an acceptable range of application for the propulsion system.

The above risk will be mitigated by careful consideration of the design requirements, in parallel with investigating concepts. By executing these activities in parallel, overambitious design requirements will be avoided.

The other important risk is that the project may exceed the allowed time duration. This risk will be mitigated by adhering to the planned time per activity. Some slack was provided in the time scales for later part of the project to provide for unexpected work or delays in completing planned activities.

The risk that the project may exceed the allowed cost is low, since the Department of Mechanical and Mechatronic Engineering already has access to the software required for the project and most of the key subsystems required for the concept demonstrator.

6. Conclusions

Electrical vehicles are increasingly gaining attention and the propulsion system is a determining factor in the performance and production cost of such a vehicle. The proposed project will identify the best propulsion system for these vehicles, and will investigate the technical feasibility of producing such a propulsion system. The results from the investigation will be verified by tests on a demonstration model. This will provide the basis for decision regarding further development and funding in electric propulsion systems.

The project team has at its disposal all the expertise required to successfully complete the project. The expected total costs are R310 100, of which R5 000 is for capital expenditure. The expected duration is 8 months. Mitigation measures are in place to address the project risks, outline above.

7. References

Formerstudent, A, 1993, "Investigation into Batteries for Electrical Vehicles ", Mechatronic Project 478 Final Report, Department Mechanical and Mechatronic Engineering, Stellenbosch University.

Appendix A: Planning Details

This appendix gives the estimated cost to complete the project in Table A.1 and a Gantt chart for all the activities in Figure A.1.

Table A.1: Estimated Cost per Activity

Activity	Engineering Time		Running Costs	Facility Use	Capital Costs	MMW Labour		MMW Material	TOTAL
	hr	R	R	R	R	hr	R	R	R
Review Literature	25	11 250	500						**11 750**
Identify Compatible Applications	25	11 250							**11 250**
Compile Design Requirements	25	11 250							**11 250**
Investigate Concepts for the Propulsion System	70	31 500	100						**31 600**
Simulate the Most Promising Concepts	100	45 000	150	1000					**46 150**
Select Preferred Concept	10	4 500							**4 500**
Design Review	5	2 250	500	500					**3 250**
Design Concept Demonstrator	50	22 500	50	200		5	1 500		**24 250**
Manufacture Concept Demonstrator	20	9 000	10 000			120	36 000	10 000	**65 000**
Test Concept Demonstrator	50	22 500	500	1 000	5 000	15	4 500		**33 500**
Validate Simulation	50	22 500							**22 500**
Finalise Report	100	45 000	100						**45 100**
TOTAL	**530**	**238 500**	**11 900**	**2 700**	**5 000**	**140**	**42 000**	**10 000**	**310 100**

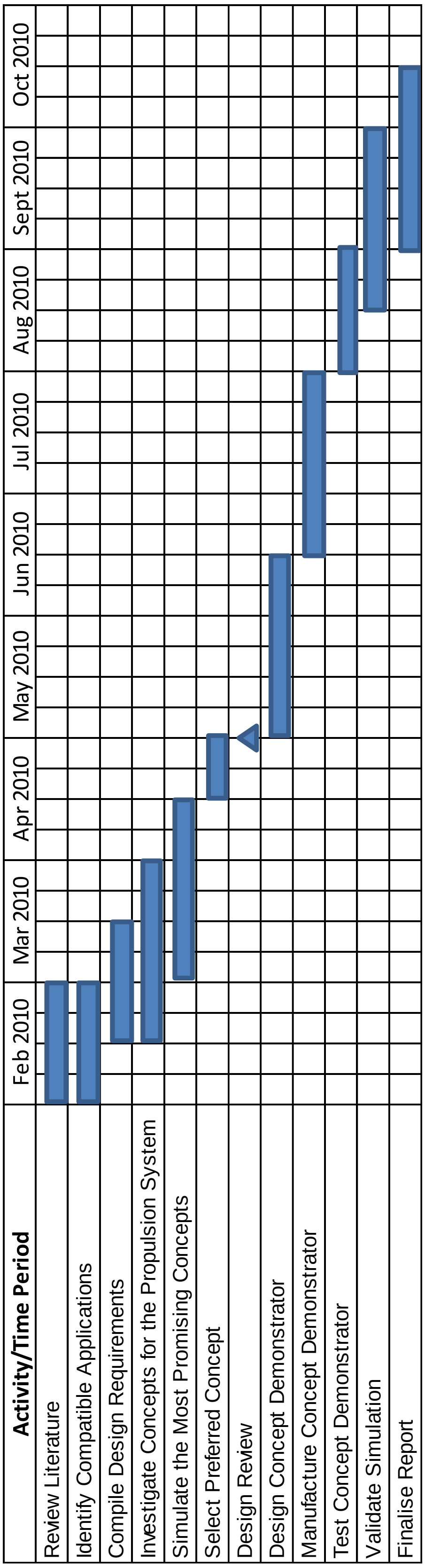

Figure A.1: Gantt Chart for Electrical Vehicle Propulsion Investigation